

LAB07_OLIVEIRA

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Laboratório 7

1. Baixe o arquivo uwmadison.sqlite3. Conecte-se a ele usando o pacote RSQLite, armazenando a conexão em uma variável conn.

```
library("RSQLite")
```

Warning: pacote 'RSQLite' foi compilado no R versão 4.5.3

```
conn <- dbConnect(SQLite(),"uwmadison.sqlite3")
dbListTables(conn)
```

```
[1] "course_offerings"  "courses"           "grade_distributions"
[4] "instructors"       "rooms"              "schedules"
[7] "sections"          "subject_memberships" "subjects"
[10] "teachings"
```

2. Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (subjects.abbreviation='STAT');

```
professores = "SELECT COUNT(DISTINCT instructors.id)
FROM instructors
JOIN teachings
  ON instructors.id = teachings.instructor_id
JOIN sections
  ON teachings.section_uuid = sections.uuid
JOIN subject_memberships
  ON sections.course_offering_uuid =
subject_memberships.course_offering_uuid
JOIN subjects"
```

```

    ON subject_memberships.subject_code = subjects.code
WHERE subjects.abbreviation='STAT' "

dbGetQuery(conn, professores)

```

```

COUNT(DISTINCT instructors.id)
1                                379

```

3. O GPA americano é definido numa escala de 0 a 4, em que A = 4, AB = 3.5, B = 3, BC = 2.5, C = 2, D = 1 e F = 0. Determinando a nota média de cada oferecimento pela ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):
- Quem é o professor mais difícil?

```

professores_d <- "
SELECT
  instructors.name,
  SUM(
    COALESCE(a_count, 0) * 4.0 +
    COALESCE(ab_count, 0) * 3.5 +
    COALESCE(b_count, 0) * 3.0 +
    COALESCE(bc_count, 0) * 2.5 +
    COALESCE(c_count, 0) * 2.0 +
    COALESCE(d_count, 0) * 1.0
  ) /
  NULLIF(SUM(
    COALESCE(a_count, 0) +
    COALESCE(ab_count, 0) +
    COALESCE(b_count, 0) +
    COALESCE(bc_count, 0) +
    COALESCE(c_count, 0) +
    COALESCE(d_count, 0) +
    COALESCE(f_count, 0)
  ), 0) AS GPA

FROM instructors
JOIN teachings
  ON instructors.id = teachings.instructor_id
JOIN sections
  ON teachings.section_uuid = sections.uuid

```

```

JOIN subject_memberships
  ON sections.course_offering_uuid = subject_memberships.course_offering_uuid
JOIN subjects
  ON subject_memberships.subject_code = subjects.code
JOIN grade_distributions
  ON grade_distributions.course_offering_uuid = sections.course_offering_uuid

WHERE subjects.abbreviation = 'STAT'

GROUP BY instructors.id, instructors.name
HAVING GPA IS NOT NULL
ORDER BY GPA ASC
LIMIT 1
"

dbGetQuery(conn, professores_d)

```

	name	GPA
1	JAMES D KUELBS	2.607692

- Quem é o professor mais fácil?

```

professores_f <- "
SELECT
  instructors.name,
  SUM(
    COALESCE(a_count, 0) * 4.0 +
    COALESCE(ab_count, 0) * 3.5 +
    COALESCE(b_count, 0) * 3.0 +
    COALESCE(bc_count, 0) * 2.5 +
    COALESCE(c_count, 0) * 2.0 +
    COALESCE(d_count, 0) * 1.0
  ) /
  NULLIF(SUM(
    COALESCE(a_count, 0) +
    COALESCE(ab_count, 0) +
    COALESCE(b_count, 0) +
    COALESCE(bc_count, 0) +
    COALESCE(c_count, 0) +
    COALESCE(d_count, 0) +
    COALESCE(f_count, 0)
  ), 0) AS GPA

```

```

FROM instructors
JOIN teachings
    ON instructors.id = teachings.instructor_id
JOIN sections
    ON teachings.section_uuid = sections.uuid
JOIN subject_memberships
    ON sections.course_offering_uuid = subject_memberships.course_offering_uuid
JOIN subjects
    ON subject_memberships.subject_code = subjects.code
JOIN grade_distributions
    ON grade_distributions.course_offering_uuid = sections.course_offering_uuid

WHERE subjects.abbreviation = 'STAT'

GROUP BY instructors.id, instructors.name
HAVING GPA IS NOT NULL
ORDER BY GPA DESC
LIMIT 1
"

dbGetQuery(conn, professores_f)

```

	name	GPA
1	SRIKANTHMADHAVAN ARAVAMUTHAN	4

- Qual é a disciplina mais difícil?

```

materia_d <- "

SELECT
courses.number,
    courses.name,
    SUM(a_count * 4 + ab_count * 3.5 +
    b_count * 3 + bc_count * 2.5 + c_count * 2 + d_count * 1)/
    SUM(a_count + ab_count + b_count +
    bc_count + c_count + d_count + f_count) AS GPA
FROM courses
JOIN course_offerings
    ON courses.uuid = course_offerings.course_uuid

JOIN subject_memberships

```

```

        ON course_offerings.uuid = subject_memberships.course_offering_uuid

JOIN subjects
    ON subject_memberships.subject_code = subjects.code

JOIN grade_distributions
    ON grade_distributions.course_offering_uuid = course_offerings.uuid

WHERE subjects.abbreviation = 'STAT'
GROUP BY courses.uuid, courses.number, courses.name
HAVING GPA IS NOT NULL
ORDER BY GPA ASC
LIMIT 1
"

dbGetQuery(conn, materia_d)

```

	number	name	GPA
1	431	Introduction to the Theory of Probability	2.886374

- Qual é a disciplina mais fácil?

```

materia_f <- "

SELECT
courses.number,
    courses.name,
    SUM(a_count * 4 + ab_count * 3.5 +
    b_count * 3 + bc_count * 2.5 + c_count * 2 + d_count * 1)/
    SUM(a_count + ab_count + b_count +
    bc_count + c_count + d_count + f_count) AS GPA
FROM courses
JOIN course_offerings
    ON courses.uuid = course_offerings.course_uuid

JOIN subject_memberships
    ON course_offerings.uuid = subject_memberships.course_offering_uuid

JOIN subjects
    ON subject_memberships.subject_code = subjects.code

```

```

JOIN grade_distributions
  ON grade_distributions.course_offering_uuid = course_offerings.uuid

WHERE subjects.abbreviation = 'STAT'
GROUP BY courses.uuid, courses.number, courses.name
HAVING GPA IS NOT NULL
ORDER BY GPA DESC
LIMIT 1
"

dbGetQuery(conn, materia_f)

```

	number		name	GPA
1	841	Nonparametric Statistics and Machine Learning Methods		4

```

Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()

```

```
[1] "2026-09-17 18:22:47 -03"
```